

TOPIC 5: CLIMATE AND WEATHER

1. Difference between Climate and Weather

Weather is the state of the atmosphere at a particular place and time. It changes from day to day or even hour to hour (e.g. sunny this morning, rainy this afternoon).

Climate is the average weather condition of a place over a long period (usually 30 years or more). It describes the general pattern of temperature, rainfall and seasons of a region (e.g. tropical climate of Nigeria).

Simple rule: Weather is short-term; Climate is long-term average.

2. Factors Affecting Weather

- **Temperature** – how hot or cold the air is
- **Humidity** – amount of water vapour in the air
- **Rainfall / Precipitation** – rain, drizzle, hail
- **Wind** – movement of air (speed and direction)
- **Cloud cover** – amount of cloud in the sky
- **Atmospheric pressure** – weight of the air
- **Sunshine / Solar radiation**

3. Measuring Weather Elements

- **Thermometer** – measures temperature (°C)
- **Rain gauge** – measures amount of rainfall (mm)
- **Wind vane / Weather vane** – shows wind direction
- **Anemometer** – measures wind speed
- **Hygrometer / Wet-and-dry bulb thermometer** – measures humidity
- **Barometer** – measures atmospheric pressure
- **Sunshine recorder** – measures duration of sunshine

4. Effects of Weather on Farming and Human Activities

On Farming:

- Rainfall determines planting and harvesting seasons.
- Too much rain can cause flooding and crop damage; too little causes drought.
- Temperature affects crop growth and animal health.
- Farmers use weather forecasts to plan activities.

On Human Activities:

- Weather affects clothing, travel, outdoor work and sports.
- Extreme weather (storms, heat waves) can disrupt schools, markets and transport.
- Health: cold and wet weather can increase respiratory illnesses; heat can cause heat stress.
- Construction, aviation and fishing are all influenced by weather conditions.

5. Summary

Weather is the day-to-day condition of the atmosphere; climate is the long-term average. Weather elements are measured with instruments such as thermometers, rain gauges and anemometers. Weather strongly affects farming, health, transport and daily life.

End of Topic 5 – Climate and Weather